Middle East Stablecoin Association

Institutional Framework Series

Proof of Reserve for Stablecoin Issuers

A Comprehensive Framework for Transparency, Risk Mitigation, and Regulatory Alignment



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Publication Notice

Proof of Reserve for Stablecoin Issuers: A Comprehensive Framework for Transparency, Risk Mitigation, and Regulatory Alignment

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MESA is a non-profit industry body that convenes and engages in the public interest across the Gulf Cooperation Council region, working with regulators, central banks, issuers, custodians, institutional investors, and corporate treasuries to advance standards for digital asset market integrity.

This document has been prepared by the Technology, Innovation and Education Committee, co-chaired by Zarah Estermann and Kristiina Lumeste, in consultation with MESA member firms and external subject-matter experts.

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Chairman's Foreword

Stablecoins have crossed a threshold. What began as a niche instrument for crypto native trading has become core infrastructure for cross-border payments, institutional settlement, tokenised markets, and corporate treasury operations. The Gulf Cooperation Council has positioned itself at the centre of this transition, with the Central Bank of the United Arab Emirates, the Virtual Assets Regulatory Authority, the Financial Services Regulatory Authority of ADGM, the Dubai Financial Services Authority, the Central Bank of Bahrain, and other regional supervisors establishing frameworks that set a credible standard for institutional quality digital money.

With that credibility comes a non-negotiable obligation: every token in circulation must, at all times, be matched by verifiable, high-quality, legally accessible reserves. The question is no longer whether Proof of Reserve should exist. The question is whether traditional practices are adequate to the scale, speed, and systemic importance that stablecoins have now acquired. In MESA's assessment, it is not.

Periodic attestation, performed once a month against a selected point in time, was designed for an earlier era. It cannot detect intraday shortfalls, bind minting logic, reconcile liabilities across multiple chains, or give supervisors the continuous visibility that systemic importance demands. The industry requires a richer architecture: independent attestation combined with decentralised oracle feeds; automated mint-and-burn enforcement at the smart contract layer; bankruptcy-remote, segregated custody; machine-readable disclosure; and, increasingly, direct supervisory access.

This framework sets out how that architecture should be constructed. It draws on the rulemaking emerging from the United States' GENIUS Act, the European Union's Markets in Crypto Assets Regulation, the Monetary Authority of Singapore's stablecoin standards, the UAE regulatory stack, and the supervisory practice developing in Hong Kong, Japan, and Switzerland.

It identifies where convergence is already occurring and where gaps remain. It is deliberately prescriptive where prescription is appropriate and deliberately principled where flexibility serves the market.

MESA publishes this document in the spirit of constructive engagement. It is intended to support issuers designing or upgrading their reserve architecture, to inform supervisors shaping implementing rules, and to equip institutional counterparties with the analytical tools required to assess counterparty risk in a market that is maturing rapidly. We welcome dialogue, critique, and improvement.

The stablecoin economy is being built in the public eye. Trust in that economy will be earned transaction by transaction, attestation by attestation, block by block. Proof of Reserve, properly designed and properly enforced, is the cornerstone on which that trust depends.



Dr. Bhaskar Dasgupta
Chairman

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Executive Summary

Stablecoins have emerged as foundational infrastructure within digital asset markets, payment systems, and, increasingly, within regulated financial services. As tokenised representations of fiat currency, they are safe and reliable only if they have transparent, continuously verifiable reserve backing. Recent regulatory developments, market failures, and institutional adoption trends converge on a single conclusion: Proof of Reserve is no longer optional. It is the foundation of credibility. Where a reserve is opaque, it transforms trust into risk.

> Proof of Reserve is no longer a disclosure exercise. It is the trust layer on which the institutional stablecoin economy will be built.
>
> MESA Technology, Innovation and Education Committee

For stablecoin issuers, Proof of Reserve frameworks, especially those incorporating real-time data feeds, independent attestations, and on-chain verification, are becoming a minimum market and regulatory standard. Regulators and policymakers globally are increasingly expecting Proof of Reserve to be part of compliance for issuers and exchanges. As the centralised and decentralised financial ecosystems become more integrated with mainstream finance, disclosure and reserve verification standards are tightening. Bank regulatory frameworks developed by the Basel Committee are expanding expectations for virtual asset exposure reporting, and some supervisory bodies have signalled intention to incorporate Proof of Reserve into routine oversight.

Despite this convergence, there is no legally binding, unified global standard for verifying the existence, composition, quality, transparency, segregation, or.

safety of reserves. Regulations across jurisdictions differ significantly. They vary on redemption timeframes, consumer protection, the balance between innovation and risk, eligible reserve assets, custodian requirements, disclosure frequency, auditing requirements, governance structures, and liability reporting. This fragmentation creates operational risk for issuers, legal uncertainty for users, unnecessary instability for financial markets, and a barrier to safe cross-border adoption.

Proof of Reserve has emerged as a foundational mechanism for improving transparency and trust across the stablecoin industry. However, it remains inconsistently defined, unevenly implemented, and often insufficient to address the full spectrum of risks associated with reserve-backed digital assets. A robust framework must encompass not only reserve verification but also liability reporting and reconciliation, governance and key management integrity, custodian segregation and bankruptcy remote structures, real time or near real time oracle verified reporting, smart contract enforced mint and burn controls, cross chain solvency synchronisation, cybersecurity hardening and oracle infrastructure resilience, prudential reserve quality and liquidity standards, and regulatory reporting pipelines with machine readable disclosures.

Why this framework, and why now

This paper presents an institutional-grade, consolidated framework for Proof of Reserve, specifically tailored for stablecoin issuers. It synthesises principles from industry practice, regulatory requirements, including the United States GENIUS Act, the European Union MiCA regime, ADGM and VARA rulebooks, MAS standards, and MESA guidance, together with technical advances such as decentralised oracle networks, Merkle tree proofs, and smart-contract-enforced minting logic.

The objective is to define a clear, academically rigorous, and operationally realistic reference model for Proof of Reserve in the stablecoin sector that supports institutional adoption, supervisory integration, and cross-jurisdictional portability.

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